Pickup Bed Side Panel — Styleside

Ford Ecat Electronic Parts Catalog

General Equipment

3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-00014
Inverter Welder with MIG Welder 254-00015

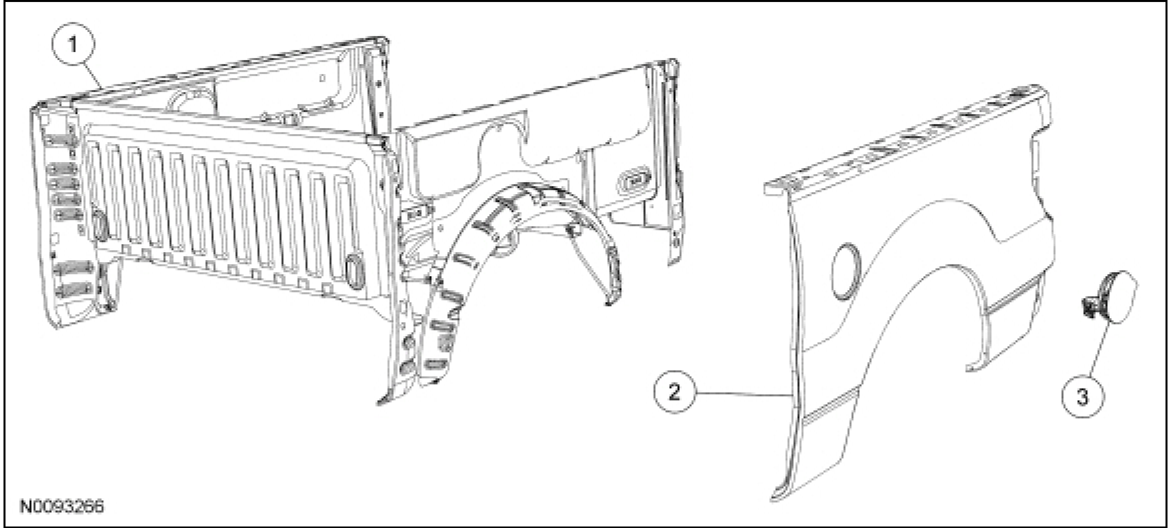
Material

Item	Specification
Flexible Foam Repair Fusor® 121	—
Motorcraft® Metal Bonding Adhesive TA-1	—
Premium Undercoating ValuGard™ VG101, VG101A (aerosol)	—
Rust Inhibitor ValuGard™ VG104, VG104A (aerosol)	—
Motorcraft® Seam Sealer TA-2	—

Pickup Bed Side Panel — Styleside

NOTE: *Factory spray in bedliner (Tough Bed®) is made of a special highly durable multi-part resin. Repairs cannot be performed using conventional body shop chemical or spray equipment. To ensure proper adhesion, correct surface texture and a lasting repair, all work must be done using products that have been tested and approved by Ford Motor Company. There are 2 repair procedure levels for repairing the Tough Bed® material. Refer to www.fordbedlinerrepair.com for further instruction.*

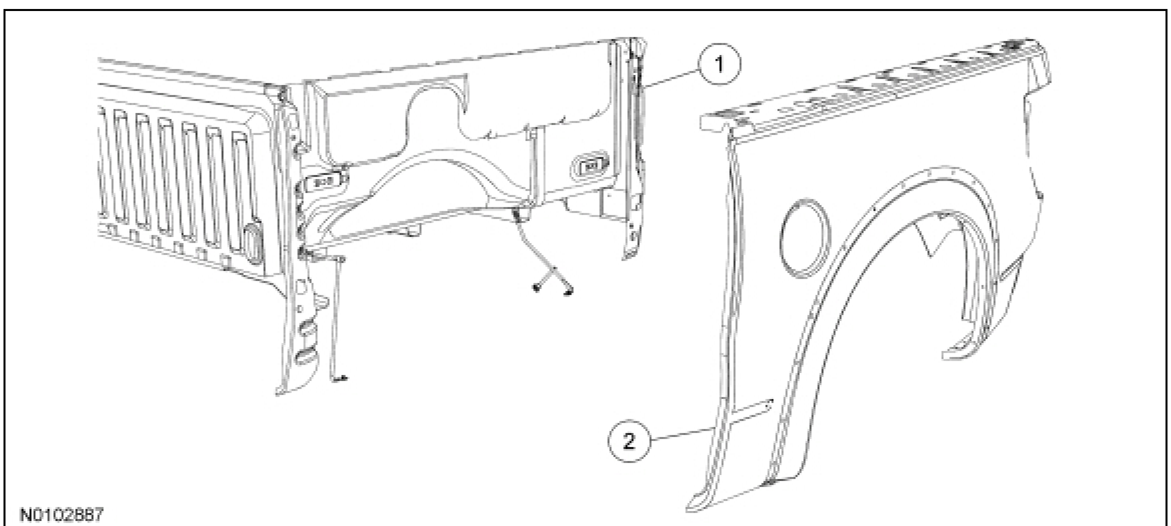
NOTE: *Left side shown, right side similar.*



Item	Part Number	Description
1	—	Pickup bed assembly — High-Strength Low Alloy (HSLA) and mild steel
2	2784127841 LH/ 27840 RH LH/ 27840 RH	Panel assembly — mild steel
3	405C46405C46	Fuel filler door assembly — mild steel

F-150 SVT Raptor

NOTE: *The Raptor outer panel is serviced as a module assembly using the same procedure as the Styleside panel.*



Item	Part Number	Description
1	—	Pickup bed assembly — High-Strength Low Alloy (HSLA) and mild steel
2	2784127841 LH/ 27840 RH LH/ 27840 RH	Panel assembly — mild steel

All Methods

WARNING: Collision damage repair must conform to the instructions contained in this workshop manual. Replacement components must be new, genuine Ford Motor Company parts. Recycled, salvaged, aftermarket or reconditioned parts (including body parts, wheels or safety restraint components) are not authorized by Ford.

Departure from the instructions provided in this manual, including alternate repair methods or the use of substitute components, risks compromising crash safety. Failure to follow these instructions may adversely affect structural integrity and crash safety performance, which could result in serious personal injury to vehicle occupants in a crash.

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

WARNING: Always refer to Material Safety Data Sheet (MSDS) when handling chemicals and wear protective equipment as directed. Examples may include but are not limited to respirators and chemically resistant gloves. Failure to follow these instructions may result in serious personal injury.

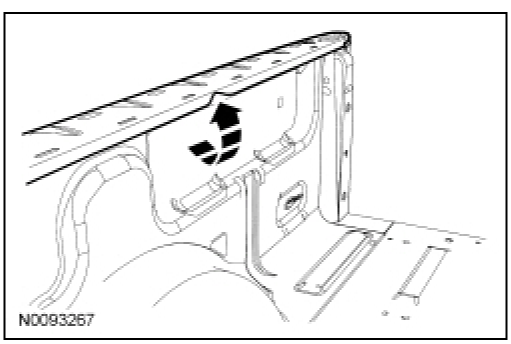
WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

NOTICE: Appropriate seam sealer needs to be applied to exposed joints and seams in addition to corrosion protection, particularly horizontal surfaces where moisture collects and migrates. Exposed seams and joints can corrode or rust prematurely if seam sealer is not applied correctly. For additional information, refer to Sealers in this section.

NOTE: *Factory spot welds may be substituted with either Squeeze-Type Resistance Spot Welding (STRW) or Metal Inert Gas (MIG) plug welds. Spot/plug welds should equal factory welds in both location and quantity. Do not place a new spot weld directly over an original weld location. Plug weld hole should equal 8 mm (0.31 in) diameter.*

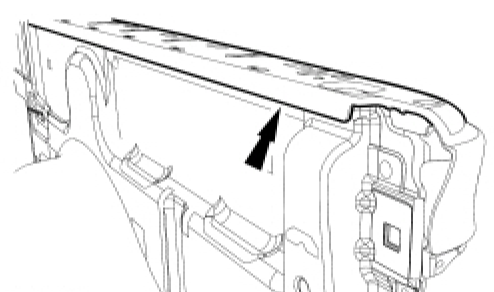
NOTE: *Observe prescribed welding procedures when carrying out any body side section repair. For additional information, refer to Welding Precautions — Steel in this section.*

1. Remove the tailgate assembly. For additional information, refer to Section 501-04.
2. Remove the rear lamp assembly from the affected side. For additional information, refer to Section 417-01.
3. If equipped, remove the wheelhouse opening moulding(s).
4. Remove the pickup bed upper rail moulding from the affected side.
5. Remove the pickup bed assembly and place on suitable workstands. For additional information, refer to Section 501-04.
6. Using suitable pliers, bend the inner flange upwards along the bed upper rail to allow access for a spot weld cutting tool.



7. **NOTICE: Use caution to only drill through the spot welds in the outer panel to avoid damage to the inner panel.**

   Using a commercially available spot weld cutter or equivalent, drill out the spot welds that are accessible.
8. Using an air chisel or hand-held weld splitter, separate the weld flanges at the bed rail and front laps of the panel.



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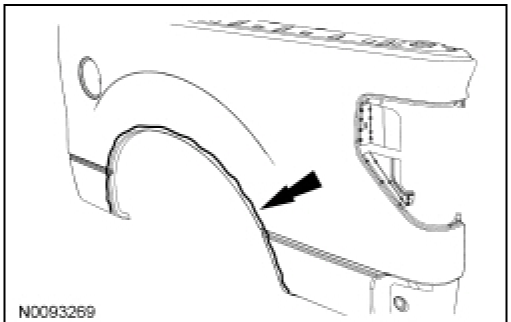
9. **NOTE:** *DO NOT remove the foam sealer from around the inside of the wheel opening*.

   Using an air saw or air chisel, cut the outer panel, making sure not to cut into the mating flanges or adjacent parts.



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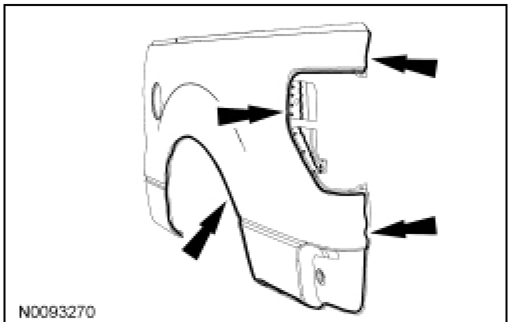
10. Continue cutting along the lines until the panel can be completely removed.



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11. Drill out the remaining spot welds.

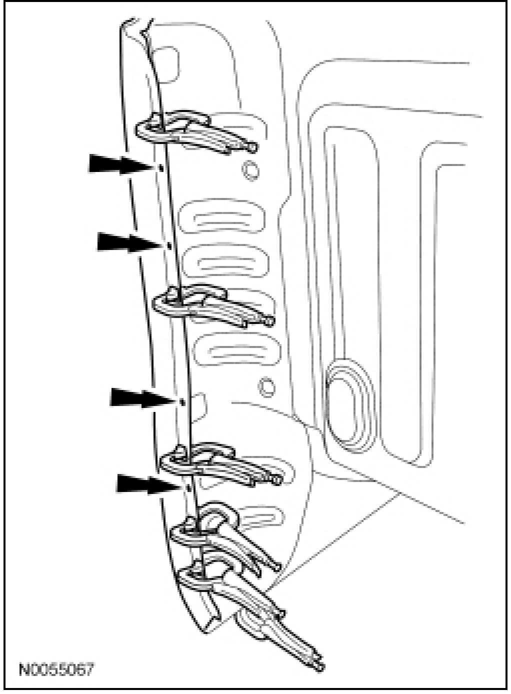
12. Using an air chisel or hand-held splitter, separate and remove the remaining outer panel material.

13. Straighten all flanges as necessary using a hammer and dolly.

14. Using a suitable grinder and grinding disc, carefully grind the mating surfaces of the receiving flanges on the vehicle as follows:
    - Make sure to remove all E-coat, paint or galvanized coating from the mating surfaces. If the metal has a pewter appearance, the galvanized coating has not been completely removed. A shiny appearance must be left after grinding.
    - DO NOT damage the corners or thin the metal.
    - DO NOT GRIND above the mating surface area. This will help eliminate the possibility of corrosion formation after the repair is completed.

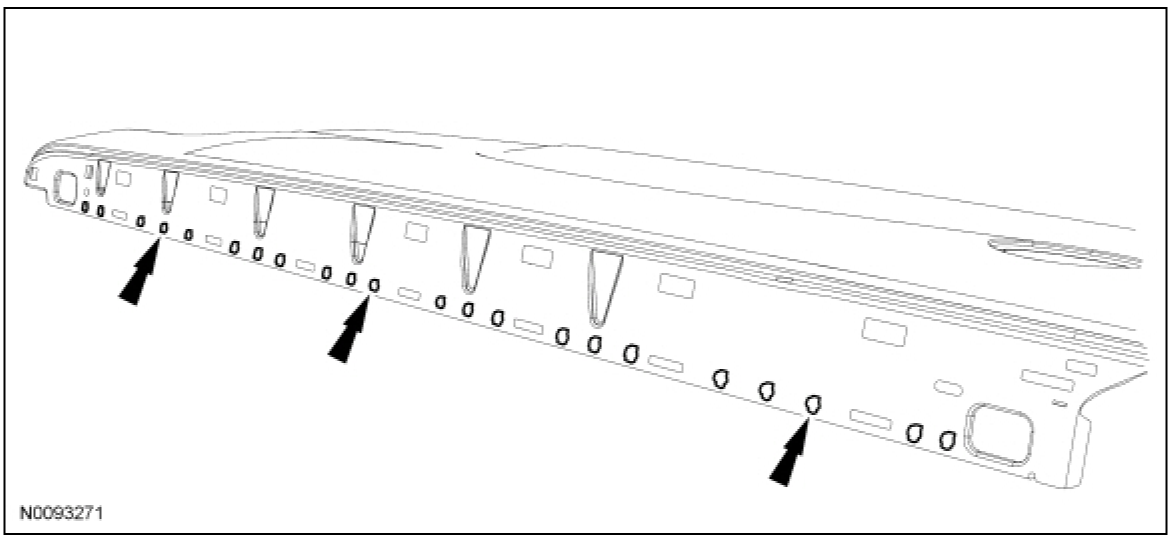
15. Dry-fit and clamp the new outer panel to check fit.

16. Using the original spot welds as a guide, mark the new spot weld locations next to, but not on top of, the original spot weld locations on the new outer panel.

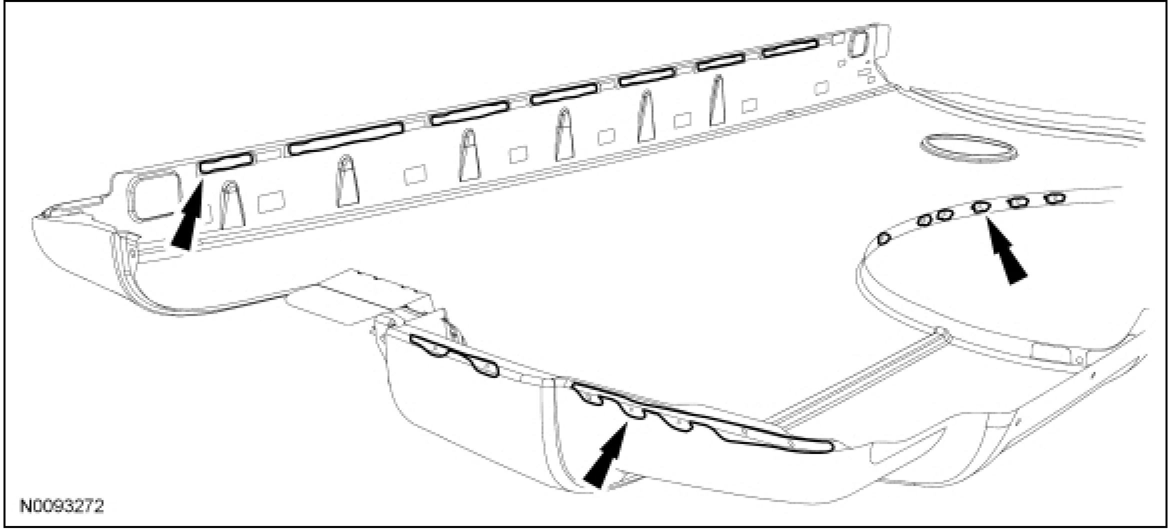


17. Remove the panel and place on suitable work stands.

Squeeze-Type Resistance Spot Welding (STRW) Method



18. Using a suitable grinder and grinding disc, carefully grind only the spots marked earlier in the new outer panel as follows:
    - Make sure to remove all E-coat, paint or galvanized coating from the mating surfaces. If the metal has a pewter appearance, the galvanized coating has not been completely removed. A shiny appearance must be left after grinding.
    - DO NOT damage the corners or thin the metal.
    - DO NOT GRIND above the mating surface area. This will help eliminate the possibility of corrosion formation after the repair is completed.



19. **NOTE:** *Set up spot welder per manufacturer's recommendations for metal thickness.*

    Prepare the metal bonding adhesive by placing the cartridge into an applicator gun and dispense a small amount of adhesive to level plungers. Install a mixing tip onto the cartridge and dispense a 75 mm (2.95 in) bead of adhesive onto a scrap piece of cardboard to make sure the components are correctly mixed and ready to apply.

20. **NOTE:** *Refer to the adhesive product label for use and preparation instructions.*

    Perform a weld bond test as follows:
    - Prepare 2 pieces of scrap for bonding. These will be used for a test weld.

21. Verify the resistance spot welder is set to manufacturer's recommended settings for the thickness of the metal.

22. Apply a 6 mm (0.23 in) to 9 mm (0.35 in) bead of adhesive onto the test pieces of metal.

23. **NOTE:** *Clamps should have tape over the ends to insulate them during welding.*

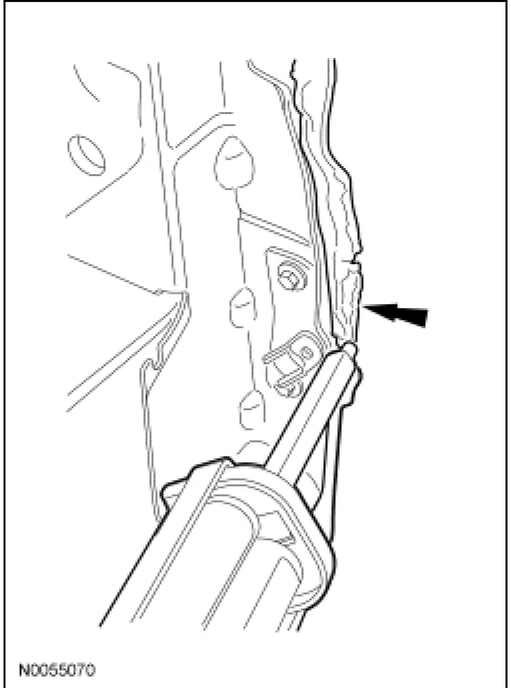
    Clamp the pieces together and apply one weld.

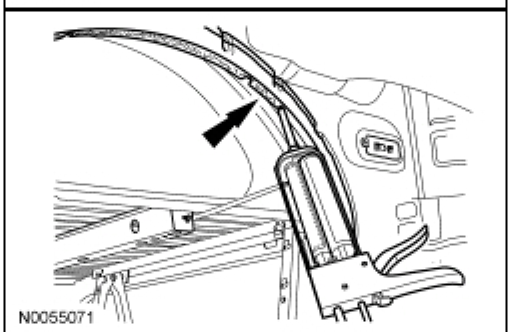
24. Remove the clamp, then place the welded pieces in a vise and perform destructive weld test by peeling the scrap metal apart using large channel lock pliers. Measure the size of the weld nugget to verify it meets Ford Motor Company weld nugget requirements. For additional information, refer to the weld nugget table in Specifications in this section.

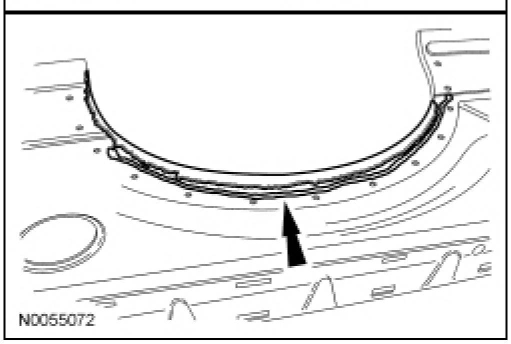
25. Proceed with the panel installation once the correct weld nugget size is achieved.

26. Wipe all dust and grit off the panel to prepare the service replacement outer panel for installation.
    - Remove any foreign material from the vehicle in the repair area.

27. Apply a bead of adhesive approximately 6 mm (0.23 in) to 10 mm (0.39 in) wide, to the flanges on the vehicle and the mating surfaces of the new outer panel in the following locations:
    - Top rail flange
    - Short vertical flange at front top of bed
    - Forward vertical flange
    - Small area at bottom of inner panel just behind forward vertical flange
    - Rear vertical flange
    - Around tail lamp area
    - Small area behind wheel opening beneath tail lamp area
    - Wheel opening flange
    - The area of the panel that contacts the wheel opening foam sealer
    - Apply an extra bead of adhesive to the inside corner around the wheel opening on the new outer panel

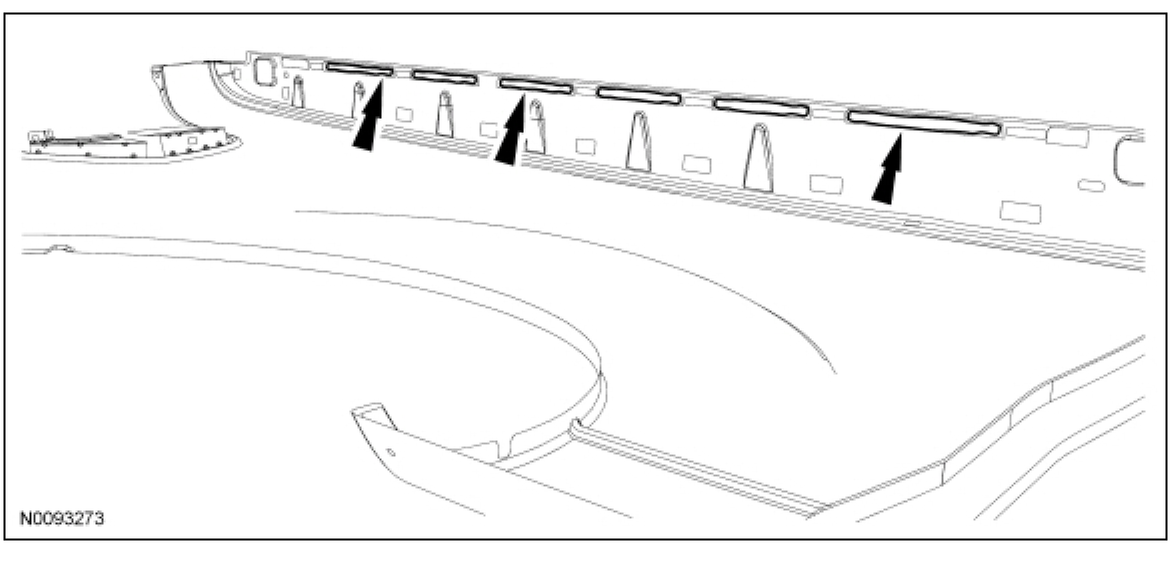






28. **NOTE:** *Do not scrape off the adhesive. Do not spread out the extra bead of adhesive that was applied to the inside corner at the wheel opening.*

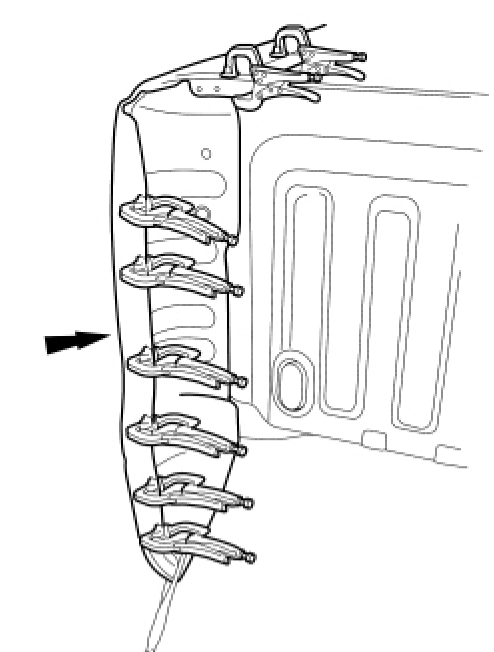
    Using a paint stick or equivalent, spread the adhesive to cover the entire area where the E-coat was ground off.



29. Position the new outer panel onto the bed. Once positioned, DO NOT PULL the panel away from the bed. If repositioning is necessary, SLIDE the panel. This maintains correct contact between the components and the adhesive.

30. **NOTE:** *Clamps should have tape over the ends to insulate them during welding.*

    Clamp the panel in place evenly and tightly. Glass beads in the adhesive will prevent over-clamping the components.



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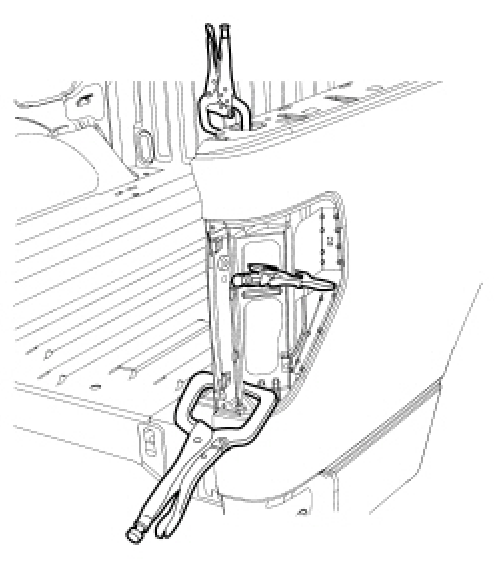

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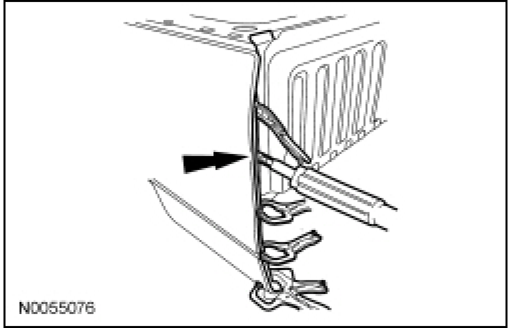
31. Wipe away any excess adhesive, where possible, before it cures.

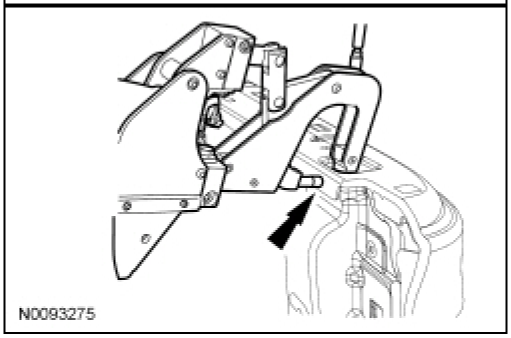
32. **NOTE:** *Welding can be carried out anytime during the adhesive curing process or after the adhesive is fully cured. Welder settings will vary when welding through wet adhesive versus welding through cured adhesive. Use the welder manufacturer's recommended settings for welding through adhesive.*

    NOTE: *It is best to place a shunt weld (in an area with no adhesive) as the first weld to make sure you have correct conductivity, particularly when welding through cured adhesive.*

    NOTE: *If welding will not be carried out immediately, follow the adhesive product label for cure times.*

    Spot weld the new panel in place, maintaining original spot weld spacing. Welds should be placed as close as possible to the original welds, BUT NOT DIRECTLY OVER AN ORIGINAL SPOT WELD.





Metal Inert Gas (MIG) Welding Method

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

WARNING: Always refer to Material Safety Data Sheet (MSDS) when handling chemicals and wear protective equipment as directed. Examples may include but are not limited to respirators and chemically resistant gloves. Failure to follow these instructions may result in serious personal injury.

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

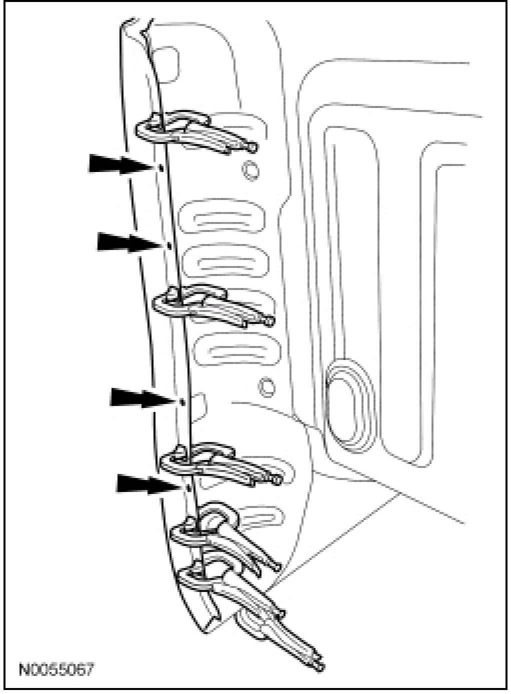
NOTICE: Appropriate seam sealer needs to be applied to exposed joints and seams in addition to corrosion protection, particularly horizontal surfaces where moisture collects and migrates. Exposed seams and joints can corrode or rust prematurely if seam sealer is not applied correctly. For additional information, refer to Sealers in this section.

33. Straighten all flanges as necessary using hammer and dolly.

34. Using a suitable grinder and grinding disc, carefully grind the mating surfaces of the receiving flanges on the vehicle as follows:
    - Make sure to remove all E-coat, paint or galvanized coating from the mating surfaces. If the metal has a pewter appearance, the galvanized coating has not been completely removed. A shiny appearance must be left after grinding.
    - DO NOT damage the corners or thin the metal.
    - DO NOT GRIND above the mating surface area. This will help eliminate the possibility of corrosion formation after the repair is completed.

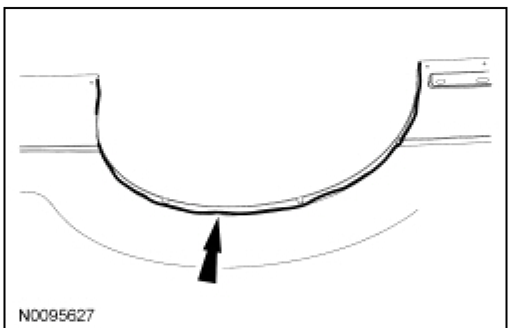
35. Dry-fit and clamp the new outer panel to check fit.

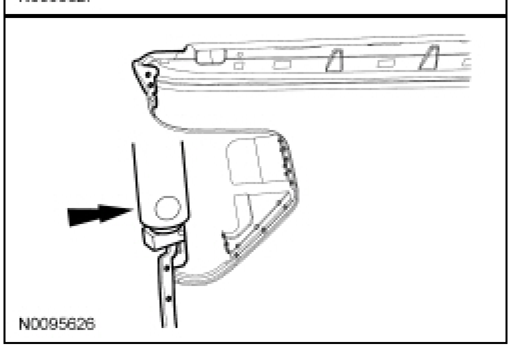
36. Using the original spot welds as a guide, mark the new spot weld locations next to, but not on top of, the original spot weld locations on the new outer panel.

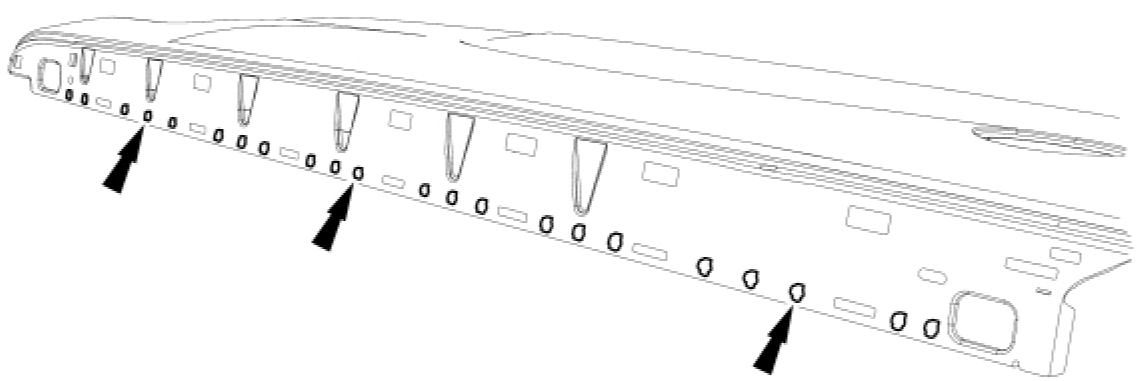


37. Remove the panel and place on suitable work stands.

38. Drill or punch the exact number of holes to match the number of spot welds that secured the original panel. The holes should be 8 mm (0.31 in) diameter and as close to the original spot welds without lining up exactly on top of the original spot welds.







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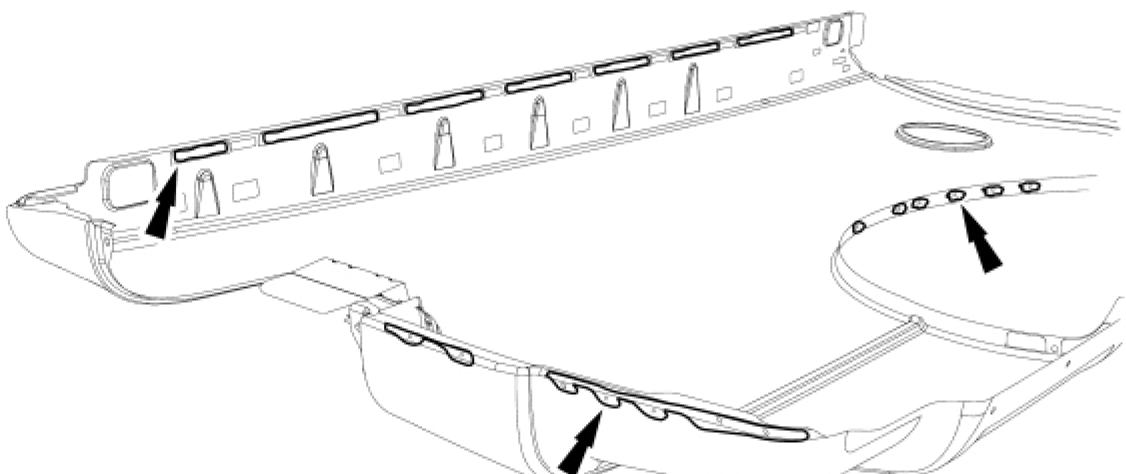
39. Using a suitable grinder and grinding disc, carefully grind only the spots marked earlier in the new outer panel as follows:
    - Make sure to remove all E-coat, paint or galvanized coating from the mating surfaces. If the metal has a pewter appearance, the galvanized coating has not been completely removed. A shiny appearance must be left after grinding.
    - DO NOT damage the corners or thin the metal.
    - DO NOT GRIND above the mating surface area. This will help eliminate the possibility of corrosion formation after the repair is completed.



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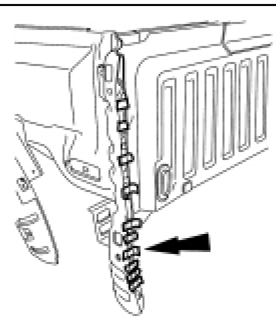
40. Wipe all dust and grit off the panel to prepare the service replacement outer panel for installation.
    - Remove any foreign material from the vehicle in the repair area.

41. Apply 25.4 mm (1 in) masking tape over the plug weld locations marked earlier to prevent contamination from the adhesive.



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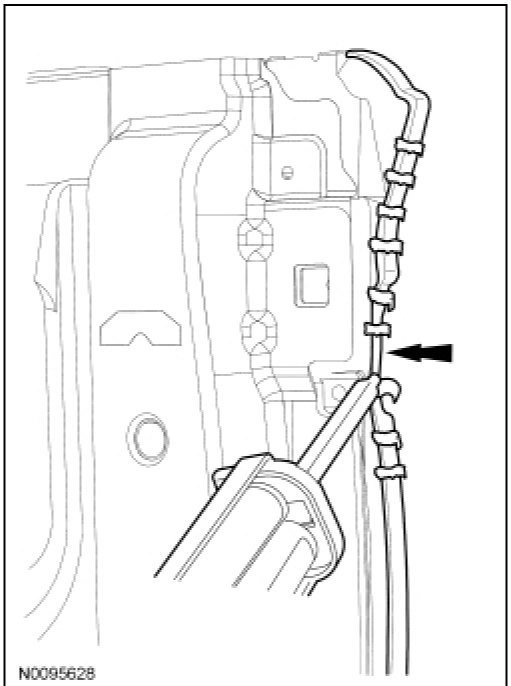
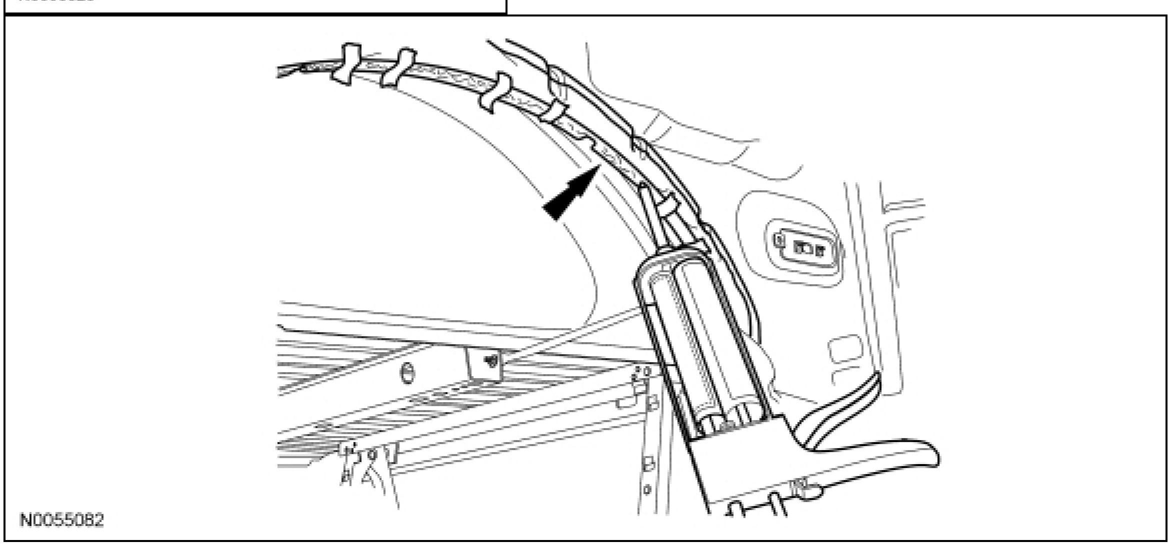
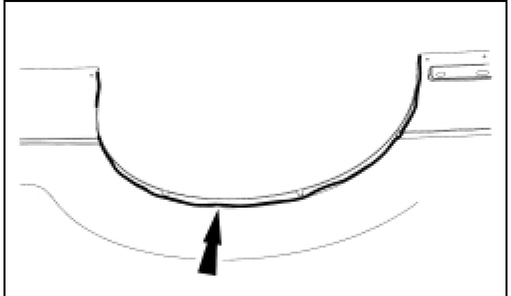
42. **NOTE:** *Refer to the adhesive product label for use and preparation instructions.*

Prepare the metal bonding adhesive by placing the cartridge into an applicator gun and dispense a small amount of adhesive to level plungers. Install a mixing tip onto the cartridge and dispense a 75 mm (2.95 in) bead of adhesive onto a scrap piece of cardboard to make sure the components are correctly mixed and ready to apply.

43. Apply a bead of adhesive approximately 6 mm (0.23 in) to 10 mm (0.39 in) wide to the flanges on the vehicle and the mating surfaces of the new outer panel in the following locations:
    - Top rail flange
    - Short vertical flange at front top of bed
    - Forward vertical flange
    - Small area at bottom of inner panel just behind forward vertical flange
    - Rear vertical flange
    - Around tail lamp area
    - Small area behind wheel opening beneath tail lamp area
    - Wheel opening flange
    - The area of the panel that contacts the wheel opening foam sealer
    - Apply an extra bead of adhesive to the inside corner around the wheel opening on the new outer panel



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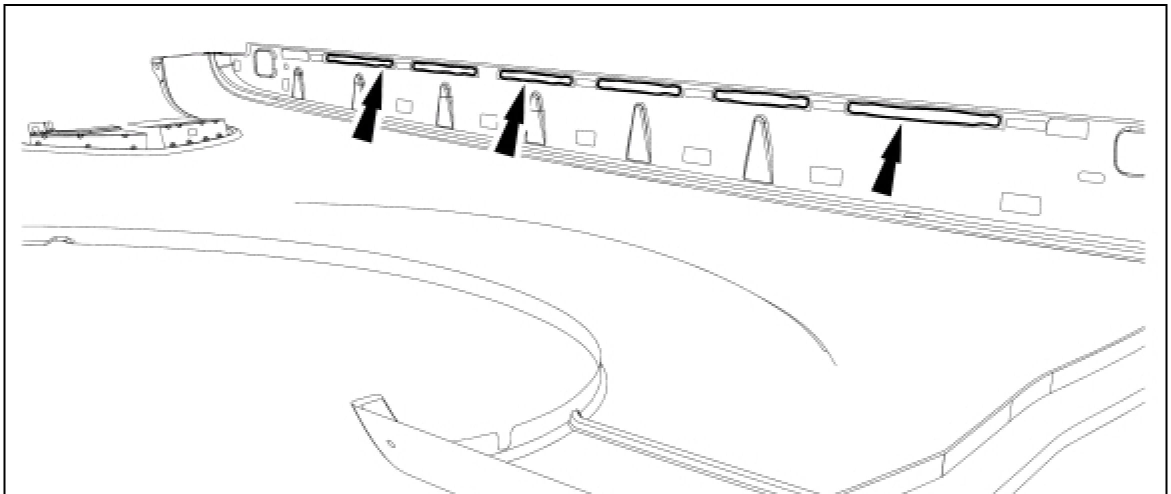
44. **NOTE:** *Do not scrape off the adhesive. Do not spread out the extra bead of adhesive that was applied to the inside corner at the wheel opening.*

    Using a paint stick or equivalent, spread the adhesive to cover the entire area where the E-coat was ground off.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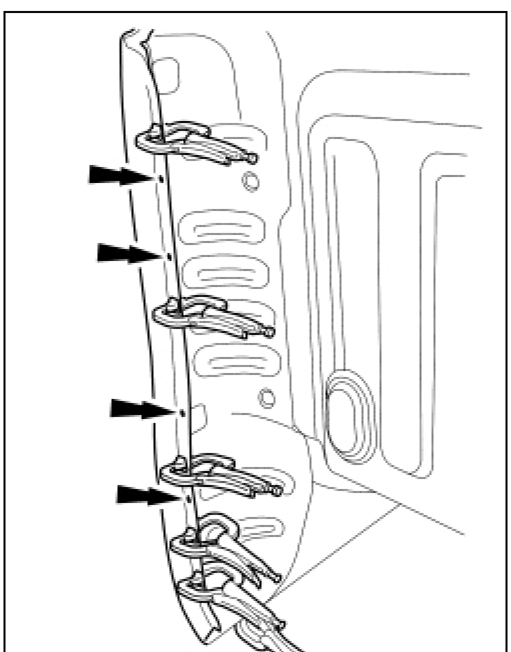
45. Remove the tape to expose the bare metal around the plug weld areas.
46. Position the new outer panel onto the bed. Once positioned, DO NOT PULL the panel away from the bed. If repositioning is necessary, SLIDE the panel. This maintains correct contact between the components and the adhesive.
47. **NOTE:** *Clamps should have tape over the ends to insulate them during welding.*

    Clamp the panel in place evenly and tightly. Glass beads in the adhesive will prevent over-clamping the components.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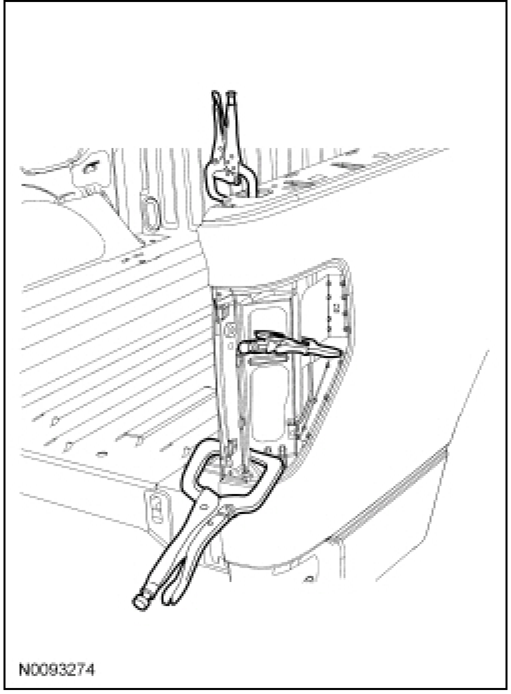
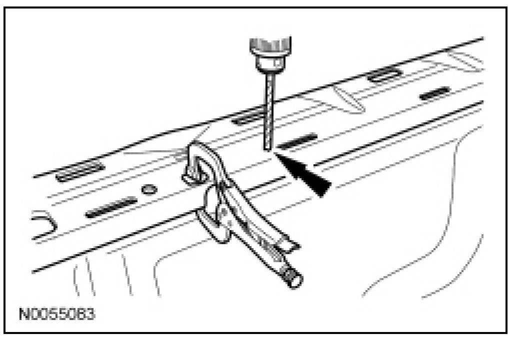
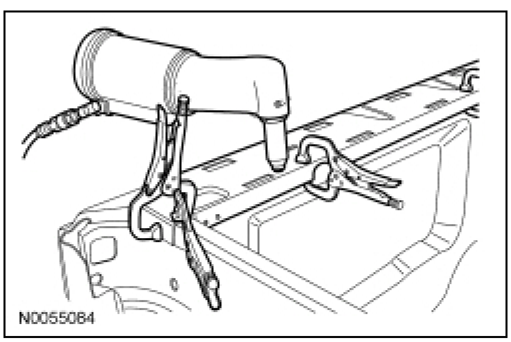


48. Using an 8 mm (0.31 in) bit, drill through the top rail in the new panel and the inner panel in the locations marked indicating the original spot welds.



49. Using locally procured rivets (3/16 x 0.440 oval head blind steel rivet), rivet the new panel in place along the upper rail.



50. **NOTE:** *Welding can be carried out anytime, while the adhesive cures or after the adhesive is fully cured. Follow the adhesive label instructions for cure times.*

    Weld the new panel in place. Welds should be placed as close as possible to the original welds, BUT NOT DIRECTLY OVER AN ORIGINAL SPOT WELD.

All methods

51. Remove the clamps after welding is completed.

52. Apply flexible foam sealer to the outer panel-to-inner panel wheel opening joint.

53. Restore corrosion protection to all bare metal surfaces. Finish any cosmetic seams with seam sealer. For additional information, refer to Restoring Corrosion Protection Following Repair in this section.

54. Proceed with the refinish process and follow Ford-approved paint procedures.

55. Reinstall the pickup bed assembly. For additional information, refer to Section 501-04.